

The figures in the right-hand margin indicate marks for the individual questions.

Answers should be concise and entire answer to a question should be together. Give examples and diagrams wherever appropriate. State assumptions wherever made.

1. Which is the primary form of a natural language- spoken form or written form? 1
2. What are the smallest elements in the written form of a language? 1
3. What are the scripts used for writing English and Hindi known as? 2
4. What is the highest frequency of human speech? What should be the sampling frequency for conversion of human speech to digital format? 1 + 1
5. Arrange the following in non-decreasing size of code-space— Unicode, ASCII and ISCII? 2
6. List the morphemes in the following- 3
boys, anybody, bat, carefully, evening
7. Suggest the lexicon entries for morphological parsing of the following words using FST- 4
cat, cats, men, talked

8. Identify cases of inflection and derivation in the following words- 3

men, slowly, writing, cats, beautiful, fish

9. What morphemes can be learnt from the following training words by an unsupervised learning approach- 3
played, playing, trying, try, careful, carefully
10. What would be the probabilities $P(\text{submit} \mid \text{to})$ and $P(\text{office} \mid \text{the})$ from the following sentences- 4

Some boys came to the office yesterday to submit their applications, but the office was closed.